Leaching of Bauxite Ore & Cementation of Copper

UMT302L: Materials Processing Laboratory · Department of Materials Engineering, IISc

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Abstract

This combined laboratory report investigates two foundational hydrometallurgical unit operations performed in the UMT302L Materials Processing Laboratory: (i) the solid–liquid extraction (leaching) of iron oxide impurities from Indian bauxite ore using 2.4 M hydrochloric acid (HCl) at 70°C, and (ii) the galvanic displacement (cementation) of dissolved copper ions (Cu^{2+}) from acidic sulfate solution (pH ≈ 2) onto a sacrificial metallic iron plate. Both extraction processes were monitored over a 90-minute reaction period by withdrawing aliquots at 15-minute intervals and quantifying elemental concentrations using flame Atomic Absorption Spectroscopy (AAS). In the bauxite leaching investigation, AAS iron standard calibration ($\lambda = 248.3$ nm, $R^2 = 0.9978$) revealed that iron dissolution occurs in two kinetic regimes: an initial rapid dissolution phase (46.33 mg/L → 52.44 mg/L in the first 15 minutes, a +13.20% increase), followed by a flat steady-state plateau at $C_{eq} \approx 52.5$ mg/L through 90 minutes, governed by product-layer diffusion limitation according to the heterogeneous Shrinking Core Model. In the copper cementation investigation, AAS copper standard calibration ($\lambda = 324.7$ nm, $R^2 = 0.9967$) recorded continuous first-order depletion from 25.96 mg/L down to 15.07 mg/L, achieving a 41.95% recovery efficiency (5.45 mg copper precipitated onto the iron plate). The reaction obeyed apparent first-order kinetics ($k_{app} = 5.92 \times 10^{-3}$ min^{-1}, $t_{1/2} = 117.1$ minutes, $R^2 = 0.9405$) under boundary-layer mass-transfer control. Comprehensive evaluations of redox thermodynamics ($\Delta E^\circ = +0.782$ V, $\Delta G^\circ = -150.9$ kJ/mol), stoichiometric mass ratios (1.138 g Cu/g Fe), chemical safety comparisons between HCl and H_2SO_4, industrial scalability, and comparative benchmarking against electrowinning are presented.

1 PART I: Leaching of Bauxite Ore for Iron Extraction

1.1 Objective

The objective of this laboratory investigation, as stated in the course syllabus handout, is to leach an Indian bauxite ore with aqueous hydrochloric acid to selectively extract iron oxide impurities into solution, and to determine the concentration of iron in the leach solution as a function of time using flame Atomic Absorption Spectroscopy (AAS).

1.2 Materials and Facilities Required

- **Raw Materials:** Finely pulverized Indian bauxite ore (2.000 g, passing 75 μm / 200-mesh sieve), high-purity alumina, and iron powder.
- **Lixiviant Solution:** Concentrated hydrochloric acid (HCl, analytical reagent grade, 35–37% w/w, sp. gr. 1.18) and deionized distilled water.
- **Reaction Apparatus:** 250 mL Pyrex borosilicate conical flask, precision micropipette (100–1000 μL) with sterile disposable polypropylene tips, 1.5 mL Eppendorf centrifuge micro-tubes, and analytical balance (±0.0001 g).
- **Thermal and Agitation Control:** Temperature-controlled orbital incubator shaker maintained isothermally at 70 ± 1°C and 180 rpm.
- **Spectro-Analytical Instrumentation:** Flame Atomic Absorption Spectrometer (Thermo Scientific / SOLAAR AAS system) equipped with an iron hollow cathode lamp (HCL) operating at resonance wavelength $\lambda = 248.3$ nm and an air–acetylene oxidizing flame ($T \approx 2300$°C).

1.3 Theoretical Background and Syllabus Questions

1.3.1 What is Leaching?

Leaching is a hydrometallurgical solid–liquid extraction unit operation in which a selective liquid solvent (lixiviant) is brought into contact with an ore, concentrate, or industrial byproduct to dissolve soluble target metals or gangue constituents selectively, leaving insoluble solid residues (tailings or leach residue) unattacked. The dissolved species pass into the liquid phase to form a pregnant leach solution (PLS), from which purified metal values are subsequently recovered by precipitation, solvent extraction, cementation, or electrowinning. In bauxite processing, hydrochloric acid selectively attacks and dissolves iron-bearing oxide minerals:

$$\mathrm{Fe_2O_3}(s) + 6\mathrm{HCl}(aq) \longrightarrow 2\mathrm{FeCl_3}(aq) + 3\mathrm{H_2O}(l) \tag{1}$$

Because bauxite is a naturally occurring mineral aggregate, actual iron-bearing phases vary between deposits (primarily hematite and goethite).

1.3.2 Common Leaching Variables

The rate of dissolution and ultimate recovery yield ($\%Y$) of a leaching process are governed by the following critical process parameters:

1. **Leaching-Agent Type and Concentration:** Higher acid activity increases the chemical driving force and accelerates protonation of surface metal–oxygen bonds.
2. **Temperature (T):** Elevating temperature increases molecular diffusivity ($D \propto T/\eta$) and exponentially accelerates interfacial reaction rates according to the Arrhenius relation ($k = k_0 \exp(-E_a/RT)$).
3. **Contact Time (t):** Governs progression along the dissolution trajectory toward thermodynamic equilibrium.
4. **Particle Size Distribution (d_p):** Finer grinding increases specific geometric surface area per unit mass ($S_w = 6/\rho_s d_p$), accelerating dissolution kinetics.
5. **Agitation Speed (N):** Vigorous stirring maintains solid suspension and thins the hydrodynamic boundary layer ($\delta \propto N^{-0.5 \text{ to } -0.8}$), reducing external liquid film mass transfer resistance.
6. **Solid-to-Liquid Ratio (Pulp Density):** Determines volumetric availability of lixiviant per gram of ore and dictates slurry rheology and mass transfer efficiency.
7. **Solution** pH **and Chemical Composition:** Controls chemical speciation, complex stability, and prevents unwanted precipitation of secondary phases.

1.3.3 Comparative Safety Precautions: HCl versus H_2SO_4

Hydrochloric acid (HCl) was used in this experiment; sulfuric acid (H_2SO_4) is also widely employed industrially. The required laboratory safety precautions differ fundamentally:

- **Hydrochloric Acid (HCl):** HCl is an aqueous solution of hydrogen chloride gas. It has a high vapor pressure and fumes intensely at ambient temperature, generating choking, corrosive vapors that attack the respiratory tract, ocular membranes, and metallic equipment. *Safety Protocol:* Must be handled strictly inside a certified chemical fume hood with acid gas respirators, neoprene gloves, and splash goggles.
- **Sulfuric Acid (H_2SO_4):** Concentrated H_2SO_4 (98%) has a very low vapor pressure, so ambient vapor inhalation is less acute. However, it is an aggressive dehydrating agent and strong oxidizing acid that rapidly chars organic tissue upon direct contact. Most crucially, mixing H_2SO_4 with water is violently exothermic ($\Delta H_{\text{dil}} \approx -74$ kJ/mol). *Safety Protocol:* **Acid must always be poured slowly into water with constant stirring, never water into acid**, to prevent localized steam generation, explosive boiling, and acid splattering. Heavy neoprene chemical aprons, full-face shields, and elbow-length butyl gloves are required.

1.3.4 Mineralogical Structure of Bauxite

Bauxite is the primary commercial ore of aluminium, consisting of hydrated aluminium oxides intimately mixed with iron, titanium, and silicon impurities:

- **Aluminous Minerals:**
  - *Gibbsite* (γ-$Al(OH)_3$ or $Al_2O_3 \cdot 3H_2O$): Monoclinic trihydrate with open layer structure bonded by hydrogen bonds; dissolves easily in caustic soda at 140–150°C.
  - *Boehmite* (γ-AlO(OH)): Orthorhombic monohydrate with layered double-sheet structure, requiring 220–240°C for digestion.
  - *Diaspore* (α-AlO(OH)): Closely packed hexagonal monohydrate, chemically refractory, requiring digestion temperatures exceeding 250°C.
- **Iron Impurities:** Iron occurs predominantly as hematite (α-Fe_2O_3, hexagonal corundum structure) and goethite (α-FeO(OH), orthorhombic structure). In acid leaching, these iron oxides dissolve according to:

$$Fe_2O_3(s) + 6H^+(aq) \longrightarrow 2Fe^{3+}(aq) + 3H_2O(l) \tag{2}$$

$$FeO(OH)(s) + 3H^+(aq) \longrightarrow Fe^{3+}(aq) + 2H_2O(l) \tag{3}$$

- **Siliceous and Titanium Gangue:** Quartz (SiO_2), kaolinite ($Al_2Si_2O_5(OH)_4$), and anatase/rutile (TiO_2).

1.3.5 Principle of Atomic Absorption Spectroscopy (AAS)

Atomic Absorption Spectroscopy is an elemental spectro-analytical technique based on the resonant absorption of optical radiation by unexcited ground-state free neutral gaseous atoms:

1. **Light Source:** A Hollow Cathode Lamp (HCL) containing an iron cathode emits sharp, discrete spectral lines corresponding to the electronic resonance transitions of iron ($\lambda = 248.3$ nm).
2. **Nebulization and Atomization:** The liquid sample solution is aspirated through a capillary, pneumatically nebulized into a fine aerosol, mixed with fuel/oxidant, and introduced into an air–acetylene flame ($T \approx 2300$°C). The solvent evaporates (desolvation), solid particles vaporize (volatilization), and molecular bonds dissociate into free neutral ground-state gaseous atoms (Fe^0).
3. **Absorption (Beer–Lambert Law):** Ground-state atoms absorb incident radiation at characteristic resonance wavelengths. Absorbance (A) is directly proportional to path length (b) and analyte concentration (C):

$$A = \log_{10}\left(\frac{I_0}{I}\right) = \varepsilon\, b\, C \tag{4}$$

   where I_0 is incident light intensity, I is transmitted intensity, and ε is molar absorptivity. For a linear calibration curve of the form $A = mC + c$, the unknown concentration is calculated by:

$$C = \frac{A - c}{m} \tag{5}$$

4. **Detection:** A grating monochromator isolates the resonant wavelength $\lambda = 248.3$ nm, and a photomultiplier tube (PMT) detects and amplifies the attenuated beam intensity.

1.4 Experimental Procedure

1. An accurately weighed 2.000 g sample of finely crushed Indian bauxite ore was transferred into a 250 mL Pyrex conical flask.
2. A 100 mL leaching acid solution consisting of 20 mL concentrated hydrochloric acid (35% HCl) and 80 mL deionized distilled water (≈ 2.4 M HCl) was prepared and added to the ore in the flask.
3. The flask was sealed with aluminium foil and placed inside an orbital incubator shaker pre-heated to 70°C and agitated at 180 rpm.

4. A 1.0 mL aliquot of the supernatant pregnant solution was withdrawn using a calibrated micropipette at intervals of $t = 0, 15, 30, 45, 60, 75$, and 90 minutes.
5. Immediately following each sampling, 1.0 mL of pre-heated deionized water was added back to maintain a constant reaction volume ($V = 100$ mL).
6. The withdrawn 1.0 mL samples were transferred into labeled Eppendorf tubes, centrifuged to separate suspended fines, diluted appropriately, and analyzed by flame AAS at $\lambda = 248.3$ nm.

1.5 AAS Calibration and Experimental Leaching Data for Iron

Table 1 displays the optical absorbance measured for standard iron calibration solutions (0–25 mg/L). Least-squares linear regression reported by the spectrometer gives:

$$A_{\mathrm{Fe}} = 0.02392 \times C_{\mathrm{Fe}}\ [\mathrm{mg/L}] + 0.0150 \quad (\mathrm{Fit}\ /\ R^2 = 0.9978) \tag{6}$$

Table 1: AAS calibration data for Iron standard solutions (Fe, $\lambda = 248.3$ nm).

Standard Level	Nominal Concentration (mg/L)	Measured Absorbance, A	Fitted Absorbance
Blank	0.0	0.000	0.015
Standard 1	5.0	0.143	0.135
Standard 2	10.0	0.260	0.254
Standard 3	15.0	0.384	0.374
Standard 4	20.0	0.495	0.493
Standard 5	25.0	0.601	0.613

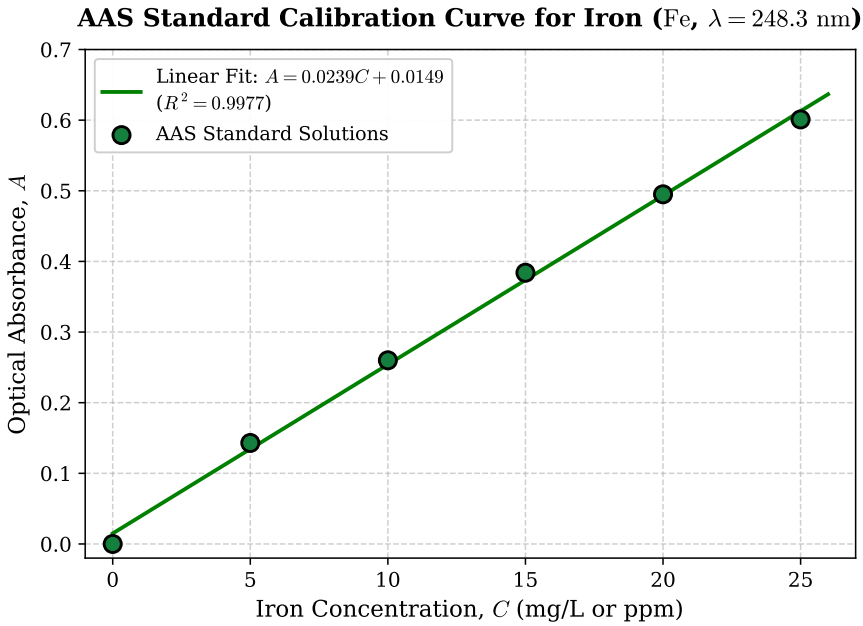


Figure 1: AAS standard calibration curve for iron (Fe, $\lambda = 248.3$ nm) showing high linear correlation ($R^2 = 0.9978$).

Table 2 presents the measured absorbance and deduced dissolved iron concentrations over the 90-minute digestion period.

Table 2: Experimental bauxite iron leaching data over time (70°C, 2.4 M HCl).

Sample No.	Time, t (min)	Measured Absorbance	Instrument Conc. (mg/L)	Corrected Conc. (mg/L)
1	0	1.123	46.3301	**46.33**
2	15	1.269	52.4441	**52.44**
3	30	1.270	52.4582	**52.46**
4	45	1.268	52.3879	**52.39**
5	60	1.268	52.3835	**52.38**
6	75	1.276	52.7466	**52.75**
7	90	1.273	52.5959	**52.60**

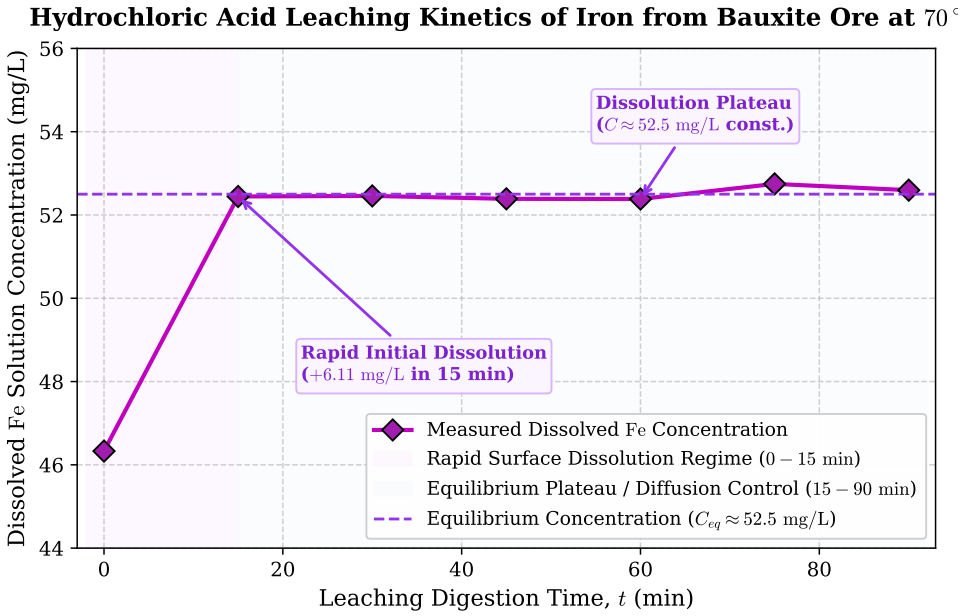


Figure 2: Iron dissolution kinetics from bauxite ore in 2.4 M HCl at 70°C, illustrating the initial rapid extraction phase (0–15 min) and subsequent steady-state equilibrium plateau (15–90 min, $C_{\text{eq}} \approx 52.5$ mg/L).

1.6 Discussion of Leaching Results and Kinetic Regimes

The experimental dissolution trajectory (Figure 2) reveals two distinct kinetic regimes:

1. **Rapid Initial Dissolution (0–15 minutes):** During the first 15 minutes, the dissolved iron concentration jumps by +6.11 mg/L (from 46.33 mg/L to 52.44 mg/L, a 13.20% increase). This rapid surge corresponds to the prompt dissolution of ultrafine, highly reactive, and unhindered iron oxide particles directly exposed on the exterior surfaces of the bauxite powder upon contact with fresh, hot acid.
2. **Equilibrium Plateau (15–90 minutes):** Between $t = 15$ min and $t = 90$ min, the iron concentration remains essentially constant (52.44 $\rightarrow$ 52.60 mg/L, within ± 0.35 mg/L instrument noise). This plateau indicates that:
   - Readily accessible surface iron oxide is completely leached within the initial 15 minutes.
   - Remaining iron oxides are encapsulated within the inert crystalline aluminosilicate/titanium matrix, or protected beneath an insoluble porous silica product layer, requiring much longer times or elevated autoclave temperatures to dissolve (shrinking core diffusion limitation).

2 PART II: Cementation of Copper Using Sacrificial Iron

2.1 Objective

The objective of this laboratory experiment, as stated in the course handout, is to remove copper from solution by the cementation process and to precipitate dissolved copper onto a sacrificial metallic iron plate, and to quantify the copper depletion profile using flame Atomic Absorption Spectroscopy (AAS).

2.2 Materials and Facilities Required

- **Reagents:** Copper(II) sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$, analytical grade, 500 mg), concentrated sulfuric acid (10 N H_2SO_4, 1–2 drops for acidifying to pH ≈ 2), and deionized water (500 mL).
- **Sacrificial Reductant:** High-purity metallic iron plate, degreased with acetone, pickled in dilute HCl to remove surface oxides, rinsed with distilled water, and dried.
- **Apparatus:** 500 mL Pyrex beaker, magnetic stirrer with Teflon stirring bar, calibrated micropipettes, and Eppendorf tubes.
- **Spectro-Analytical Instrumentation:** Flame AAS equipped with a copper hollow cathode lamp operating at resonance wavelength $\lambda = 324.7$ nm with an oxidizing air–acetylene flame.

2.3 Theoretical Background and Syllabus Questions

2.3.1 Where are Cu and Fe Relatively on the Electrochemical Series Chart?

In the standard electrochemical reduction potential series (25°C, 1 atm, 1.0 M activity):

$$\mathrm{Fe}^{2+}(aq) + 2e^- \rightleftharpoons \mathrm{Fe}(s) \qquad E^\circ = -0.440 \text{ V} \tag{7}$$

$$\mathrm{Cu}^{2+}(aq) + 2e^- \rightleftharpoons \mathrm{Cu}(s) \qquad E^\circ = +0.342 \text{ V} \tag{8}$$

Iron has a much more negative standard reduction potential (-0.440 V) than copper ($+0.342$ V). Therefore:

- In a standard reduction potential table, iron is positioned **lower** than copper (more negative potential).
- In terms of chemical reactivity (the galvanic / electromotive series of metals), iron is **higher** and more electrochemically active (anodic) than copper.
- Consequently, metallic iron (Fe^0) serves as a powerful thermodynamic reducing agent capable of transferring electrons spontaneously to reduce cupric ions (Cu^{2+}) out of solution.

2.3.2 Redox Cementation Reaction

The galvanic displacement reaction proceeds spontaneously at the iron–solution interface:

$$\mathrm{Fe}(s) + \mathrm{Cu}^{2+}(aq) \longrightarrow \mathrm{Fe}^{2+}(aq) + \mathrm{Cu}(s) \downarrow \tag{9}$$

The standard electromotive force (ΔE°) and Gibbs free energy (ΔG°) are:

$$\Delta E^\circ = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = +0.342 \text{ V} - (-0.440 \text{ V}) = +\mathbf{0.782} \text{ V} \tag{10}$$

$$\Delta G^\circ = -nF\Delta E^\circ = -2 \times 96{,}485 \text{ C/mol} \times 0.782 \text{ V} = -\mathbf{150.9} \text{ kJ/mol} \tag{11}$$

The large positive cell voltage ($+0.782$ V) and strongly negative free energy confirm that cementation is intensely spontaneous, driving copper precipitation to near-total completion ($[Cu^{2+}]_{\text{eq}} \approx 10^{-27}$ M).

2.3.3 Stoichiometry and Parasitic Side Reactions

From Equation (9), the theoretical stoichiometric ratio is 1.0 mol Cu : 1.0 mol Fe:

$$\text{Theoretical Ratio} = \frac{M_{\text{Cu}}}{M_{\text{Fe}}} = \frac{63.546 \text{ g/mol}}{55.845 \text{ g/mol}} = \mathbf{1.138} \text{ g Cu precipitated / g Fe consumed} \tag{12}$$

In practical industrial hydrometallurgy, actual iron consumption exceeds stoichiometry (1.2–2.5 g Fe/g Cu) due to parasitic side reactions:

1. **Acid Corrosion (Hydrogen Evolution):**

$$\text{Fe}(s) + 2\text{H}^+(aq) \longrightarrow \text{Fe}^{2+}(aq) + \text{H}_2(g) \uparrow \tag{13}$$

2. **Ferric Ion Reduction:**

$$\text{Fe}(s) + 2\text{Fe}^{3+}(aq) \longrightarrow 3\text{Fe}^{2+}(aq) \tag{14}$$

3. **Dissolved Oxygen Corrosion:**

$$2\text{Fe}(s) + \text{O}_2(aq) + 4\text{H}^+(aq) \longrightarrow 2\text{Fe}^{2+}(aq) + 2\text{H}_2\text{O}(l) \tag{15}$$

2.4 Experimental Procedure

1. Exactly 500 mL of deionized water was poured into a 500 mL beaker.
2. 1–2 drops of 10 N H_2SO_4 were added to adjust the solution to pH ≈ 2 (preventing premature copper hydrolysis), followed by dissolution of 500 mg $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.
3. A Teflon-coated magnetic stirring bar was placed inside, and uniform agitation was established.
4. A clean, pickled iron plate was inserted vertically into the solution at $t = 0$ min.
5. A 1.0 mL aliquot of the solution was withdrawn every 15 minutes ($t = 0, 15, 30, 45, 60, 75, 90$ min).
6. Each withdrawal was immediately replenished with 1.0 mL of deionized water to preserve constant solution volume ($V = 500$ mL).
7. Aliquots were diluted with pH 2 acidified water and analyzed by flame AAS at $\lambda = 324.7$ nm.

2.5 AAS Calibration and Experimental Cementation Data for Copper

Table 3 lists the absorbance measured for standard copper solutions (0–25 mg/L). Least-squares regression gives:

$$A_{\text{Cu}} = 0.03389 \times C_{\text{Cu}} \text{ [mg/L]} + 0.02852 \quad (R^2 = 0.9967) \tag{16}$$

Table 3: AAS calibration data for Copper (Cu, $\lambda = 324.7$ nm).

Standard Level	Nominal Concentration (mg/L)	Measured Absorbance, A	Fitted Absorbance
Blank	0.0	0.006	0.029
Standard 1	5.0	0.207	0.198
Standard 2	10.0	0.379	0.367
Standard 3	15.0	0.553	0.537
Standard 4	20.0	0.716	0.706
Standard 5	25.0	0.852	0.876

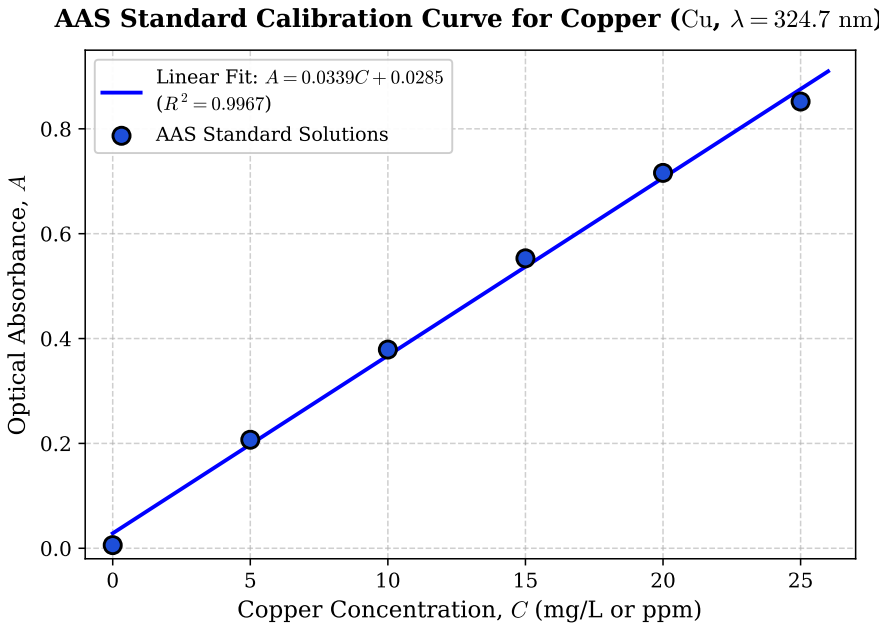


Figure 3: AAS standard calibration curve for copper (Cu, $\lambda = 324.7$ nm) showing excellent linearity ($R^2 = 0.9967$).

Table 4 provides the measured absorbance, copper concentration, and fraction of copper removed over 90 minutes.

Table 4: Copper cementation kinetic data over time on sacrificial iron plate ($V = 500$ mL, pH ≈ 2).

Sample No.	Time, t (min)	Absorbance	Conc., C_t (mg/L)	$\ln(C_t/C_0)$	Cu Removed (%)
1	0	0.908	25.9647	0.0000	**0.00%**
2	15	0.787	22.3907	-0.1481	**13.76%**
3	30	0.825	23.5229	-0.0988	**9.40%**
4	45	0.766	21.7799	-0.1758	**16.12%**
5	60	0.707	20.0227	-0.2599	**22.88%**
6	75	0.596	16.7485	-0.4384	**35.49%**
7	90	0.539	15.0740	-0.5437	**41.95%**

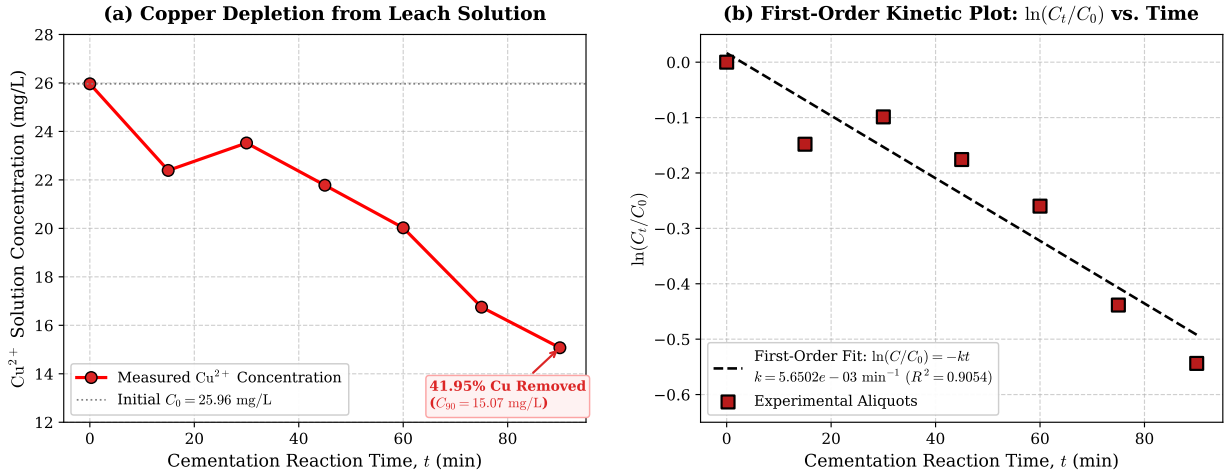


Figure 4: Copper cementation kinetics: (a) Continuous depletion of dissolved Cu^{2+} ions from 25.96 mg/L down to 15.07 mg/L (41.95% total recovery); (b) First-order kinetic model $\ln(C_t/C_0)$ versus time ($k = 5.92\times10^{-3}$ min^{-1}, $R^2 = 0.9405$).

2.6 Kinetic Modeling and Rate Analysis

Cementation of copper onto iron is widely accepted to follow first-order mass-transfer limited kinetics:

$$-\frac{\mathrm{d}C}{\mathrm{d}t} = k\left(\frac{A}{V}\right)C = k_{\mathrm{app}}C \implies \ln\left(\frac{C_t}{C_0}\right) = -k_{\mathrm{app}}t \tag{17}$$

where k is the mass transfer coefficient (cm/min), A is active iron plate area (cm^2), V is solution volume (500 mL), and k_{app} is the apparent rate constant (min^{-1}). From the slope of the linear fit in Figure 4b:

$$k_{\mathrm{app}} = \mathbf{5.92\times10^{-3}}\ \mathrm{min^{-1}} = \mathbf{3.55\times10^{-4}}\ \mathrm{s^{-1}} \quad (R^2 = 0.9405) \tag{18}$$

The half-life of copper in the reactor is:

$$t_{1/2} = \frac{\ln 2}{k_{\mathrm{app}}} = \frac{0.69315}{5.92\times10^{-3}\ \mathrm{min^{-1}}} \approx \mathbf{117.1}\ \text{minutes} \tag{19}$$

2.7 Amount of Copper Reduced and Iron Consumed

Over the 90-minute run, the concentration of copper decreased by $\Delta C = 25.9647 - 15.0740 = 10.8907$ mg/L. In the 500 mL (0.500 L) bath:

$$\text{Total Copper Precipitated: } \Delta m_{\mathrm{Cu}} = 10.8907\ \mathrm{mg/L}\times 0.500\ \mathrm{L} = \mathbf{5.445}\ \mathrm{mg} = \mathbf{5.45\times10^{-3}}\ \mathrm{g} \tag{20}$$

$$\text{Theoretical Iron Consumed: } \Delta m_{\mathrm{Fe,theo}} = \frac{\Delta m_{\mathrm{Cu}}}{1.138} = \frac{5.445\ \mathrm{mg}}{1.138} = \mathbf{4.785}\ \mathrm{mg} = \mathbf{4.79\times10^{-3}}\ \mathrm{g} \tag{21}$$

2.8 Results, Course Discussion, and Comparison Questions

2.8.1 Comments on Experimental Graphs and Analytical Trends

- **Calibration Curve (Absorbance vs. Concentration):** Figure 3 demonstrates a rigorous linear relationship ($R^2 = 0.9967$) verifying Beer–Lambert's law across the 0–25 mg/L range, indicating accurate preparation of standard solutions and proper optical alignment of the spectrometer.
- **Concentration vs. Time:** Figure 4a shows that the copper concentration in the solution decreases steadily and continuously over the 90-minute period. A dense, characteristic reddish-brown metallic copper film was visibly observed depositing onto the iron plate, confirming successful galvanic precipitation.

2.8.2 Industrial Scalability and Practical Operating Scales

Cementation is commercially practiced on an **industrial, large-scale basis**:

1. **Dilute Heap/Dump Leach Solutions (**0.5–5.0 g/L Cu**):** Where copper concentrations are too low for direct electrowinning to operate with high current efficiency.
2. **Effluent Polishing and Environmental Remediation:** Stripping heavy metals (Cu, Cd, Pb, Ni) from acid mine drainage and electroplating wastewater to regulatory discharge thresholds (< 1 ppm).
3. **Industrial Scavenging Cells:** Rotating contactor drums (Kenney cones) charged with inexpensive shredded post-consumer scrap tin-plated steel cans.

2.8.3 Comparative Evaluation with Electrowinning (Lab 5)

Table 5 presents a comprehensive comparative evaluation of Cementation against Electrowinning for copper extraction.

Table 5: Comprehensive comparison of Cementation versus Electrowinning for copper recovery.

Parameter	Cementation (Chemical Reduction)	Electrowinning (Electrochemical Reduction)
Operating Principle	Spontaneous galvanic displacement ($Fe^0 \rightarrow Fe^{2+}$).	Non-spontaneous electrolysis forced by external DC power.
Energy Requirement	Zero electrical power; driven by chemical affinity ($\Delta G^\circ < 0$).	High electricity consumption: 2.0–2.5 kWh/kg Cu ($E_{\text{cell}} \approx 2.0$ V).
Feed Concentration	Effective at very low concentrations ($<$ 1–5 g/L Cu).	Requires concentrated electrolytes (30–50 g/L Cu) to prevent mass transfer starvation.
Product Quality	Low purity ("cement copper", 80–90% Cu), contaminated with Fe and oxides. Requires subsequent smelting.	High purity (99.99% Cu, LME Grade A cathode) ready for direct commercial market.
Reagent Cost	Consumes scrap iron (1.2–2.5 t Fe/t Cu).	Uses inert anodes (Pb–Ca–Sn or Ti/IrO_2); regenerates H_2SO_4.
Environmental Impact	Discharges massive quantities of dissolved iron ($FeSO_4$) requiring neutralization.	Closed-loop acid recycling; oxygen evolution at anode.

3 Laboratory Safety and Experimental Precautions

1. **Acid Handling and Fume Protection:** Hydrochloric acid (HCl) produces highly corrosive, choking vapors and must be measured and added strictly inside a certified chemical fume hood. Wear personal protective equipment including splash goggles, chemical-resistant nitrile/neoprene gloves, and lab coat.
2. **Acid Dilution Rule:** When preparing aqueous acid solutions, always add acid slowly into water with continuous stirring; never add water to concentrated acid.
3. **Thermal and Agitation Control:** Maintain the incubator shaker strictly at $70 \pm 1^\circ$C and agitation at 180 rpm to ensure uniform mass transfer without splashing.
4. **Volume Replenishment:** Promptly replace each withdrawn 1.0 mL aliquot with exactly 1.0 mL of deionized water to prevent evaporative concentration drift and maintain $V_{\text{total}} = 100$ mL (leaching) and 500 mL (cementation).
5. **Contamination Prevention:** Use fresh, sterile micropipette tips for every withdrawal and standard aspiration to avoid cross-contamination.
6. **AAS Operating Range:** Ensure all sample absorbances fall strictly within the validated linear dynamic range of the spectrometer (0–25 mg/L); over-range samples must be quantitatively diluted.

4 Combined Conclusions

1. **Leaching of Bauxite Ore:**
   - Dissolution of iron oxide from bauxite in 2.4 M HCl at 70°C was quantitatively tracked via flame AAS ($\lambda = 248.3$ nm, calibration $R^2 = 0.9978$).
   - Iron concentration increased from 46.33 mg/L to 52.44 mg/L within the first 15 minutes (+13.20%), and remained in an asymptotic steady-state plateau at $C_{\text{eq}} \approx 52.5$ mg/L through 90 minutes.
   - The dissolution behavior reflects rapid initial attack on reactive surface oxides, transitioning to diffusion-limited mass transport through the inert aluminosilicate leach residue.
2. **Cementation of Copper:**
   - Galvanic displacement of Cu^{2+} onto sacrificial iron is intensely spontaneous ($\Delta E^\circ = +0.782$ V, $\Delta G^\circ = -150.9$ kJ/mol).
   - Copper concentration in the 500 mL bath decreased from 25.96 mg/L to 15.07 mg/L in 90 minutes, achieving a 41.95% recovery yield (5.45 mg Cu precipitated, corresponding to 4.79 mg Fe consumed theoretically).
   - The reaction obeys apparent first-order kinetics with $k_{\text{app}} = 5.92 \times 10^{-3}$ min^{-1} (3.55×10^{-4} s^{-1}, $R^2 = 0.9405$) and a reaction half-life of $t_{1/2} \approx 117.1$ minutes, governed by aqueous boundary-layer mass transfer.
3. **Hydrometallurgical Flowsheet Integration:** Leaching generates the metal-bearing pregnant solution, while cementation provides a low-cost, electricity-free extraction route for dilute streams, ideally complementing electrowinning in integrated mineral processing circuits.

References

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